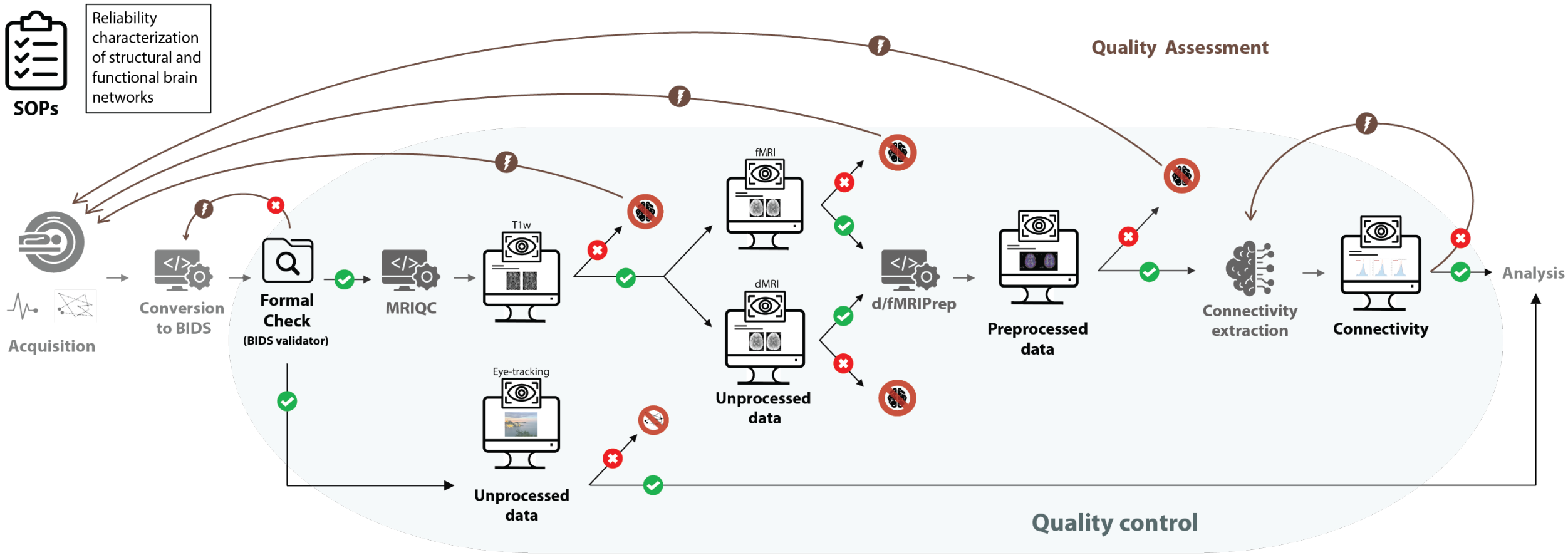

SOPs
Reliability characterization of structural and functional brain networks
Quality Assessment
Acquisition
Conversion to BIDS
Formal Check
(BIDS validator)
MRIQC
T1w
fMRI
dMRI
Unprocessed data
d/fMRIPrep
Preprocessed data
Connectivity extraction
Connectivity
Analysis
Eye-tracking
Unprocessed data
Quality control

QA/QC of Anatomical, Functional, and Diffusion MRI Data of the Human Connectome Phantom (HCPh) Dataset with MRIQC and fMRIPrep

Céline Provins[1] and Oscar Esteban[1]

[1]Department of Radiology, Lausanne University Hospital & University of Lausanne, Switzerland

ISMRM 2024

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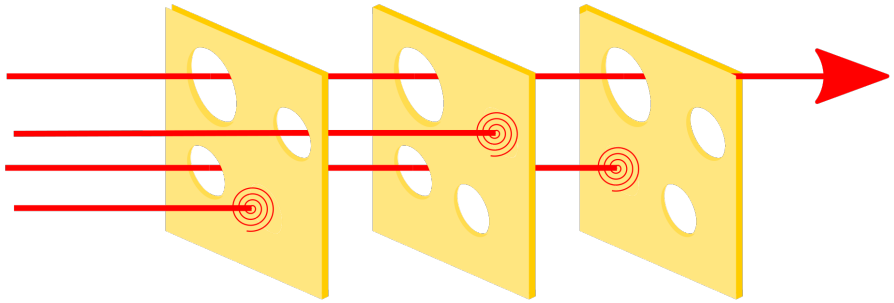
Introduction

- Data of **insufficient quality** introduce **biases** across MRI modalities [1-3].
- Robust quality assessment and quality control (QA/QC) is **key to reliable** neuroimaging analyses [4,5].
- **Visual** verification remains vital to **understand** the data [6,7] as well as to spot and interpret **unexpected** issues [8].

[1] Power et al. 2012, NeuroImage
[2] Yendiki et al. 2014, NeuroImage
[3] Reuter et al. 2015, NeuroImage
[4] Niso et al. 2022, NeuroImage
[5] Taylor et al. 2023, NeuroImage
[6] Taylor et al. 2023, Front. Neurosci.
[7] Teves et al. 2023, Front. Neurosci.
[8] Di et al. 2023, Front. Neuroimaging

Introduction

- We conceptualize QA/QC as a Swiss cheese security model, that is we set up **multiple QC checkpoints** along the neuroimaging pipeline to prevent subpar images from reaching analysis [9].



https://en.wikipedia.org/wiki/Swiss_cheese_model

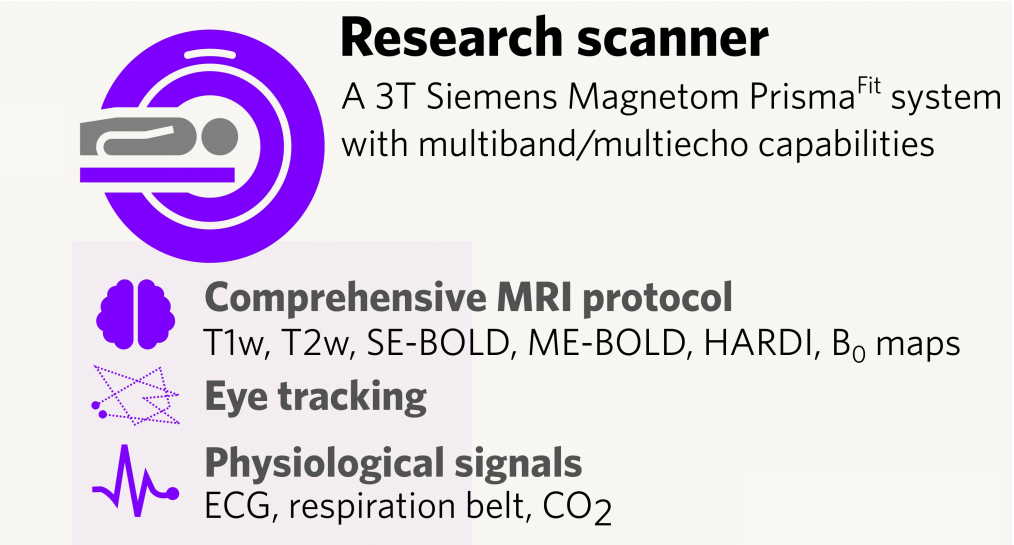
- Here, we present the QA/QC protocol for our mid-sized, single-subject acquisition effort.

[9] Provins et al. 2023, Front. Neuroimaging

Methods

Data

59 sessions from the Human Connectome Phantom (HCPh) project [10]:

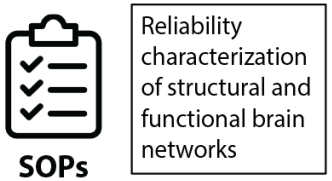


[10] Provins et al. 2023, Nat. Methods

Formalization of Exclusion Criteria

- The HCPh project is accompanied by comprehensive **Standard Operating Procedures (SOPs)** [11] documenting in detail all aspects of the experiment settings (publicly available online; https://www.axonlab.org/hcph-sops)

[11] Provins et al. 2023, Online Rep.

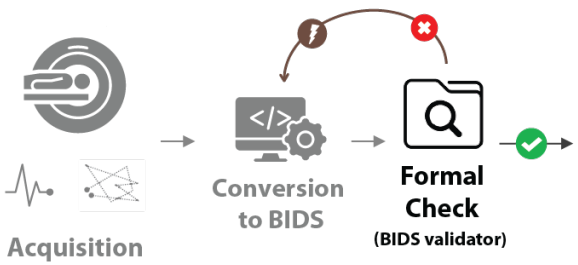


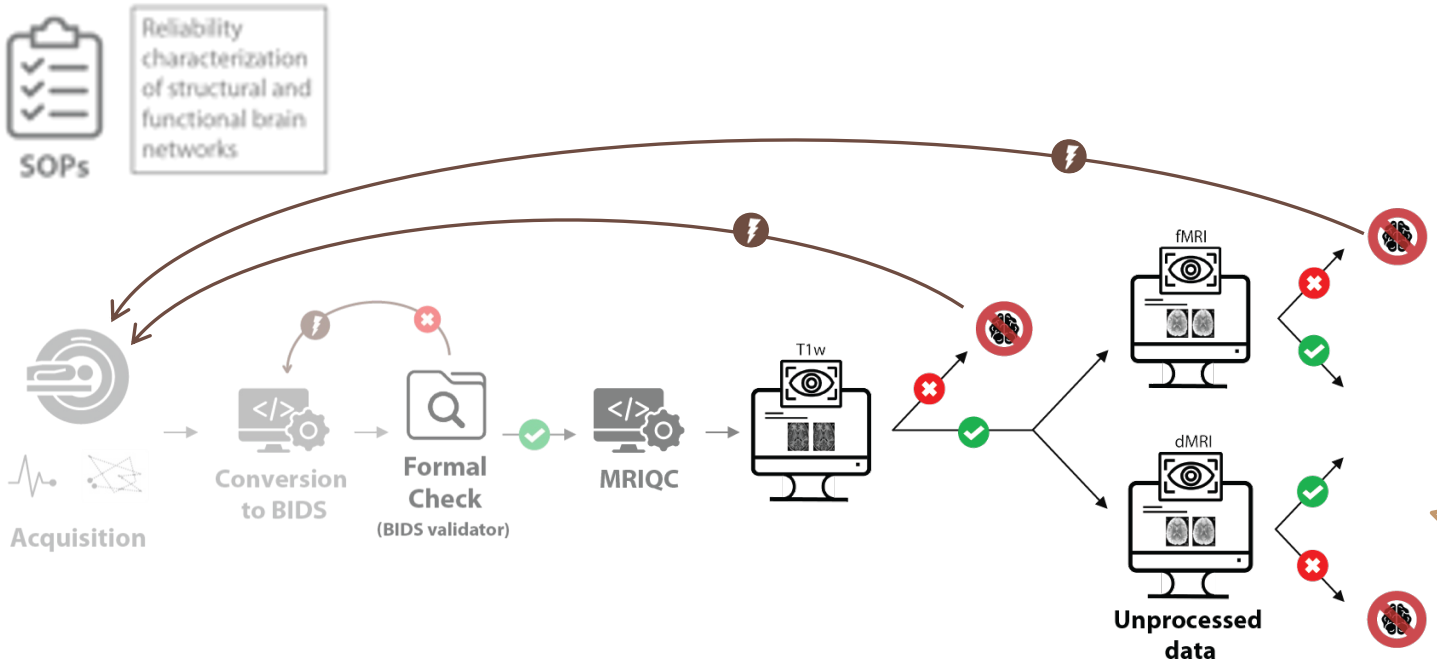
Reliability characterization of structural and functional brain networks

- Each QC checkpoint comes with exclusion criteria **tailored** to the specific application, goals, and data availability of the HCPh study.



Quality Assessment





Quality Assessment

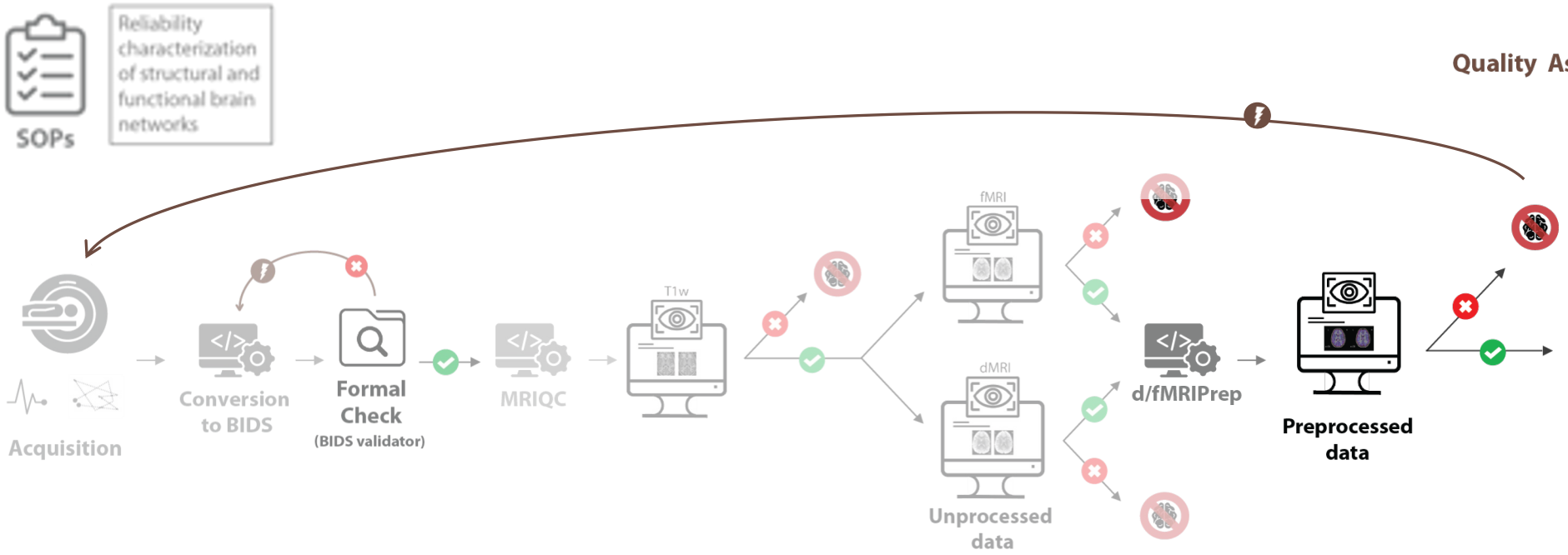
Harnessing QA/QC protocols for diffusion MRI neuroimaging workflows with MRIQC

#0539

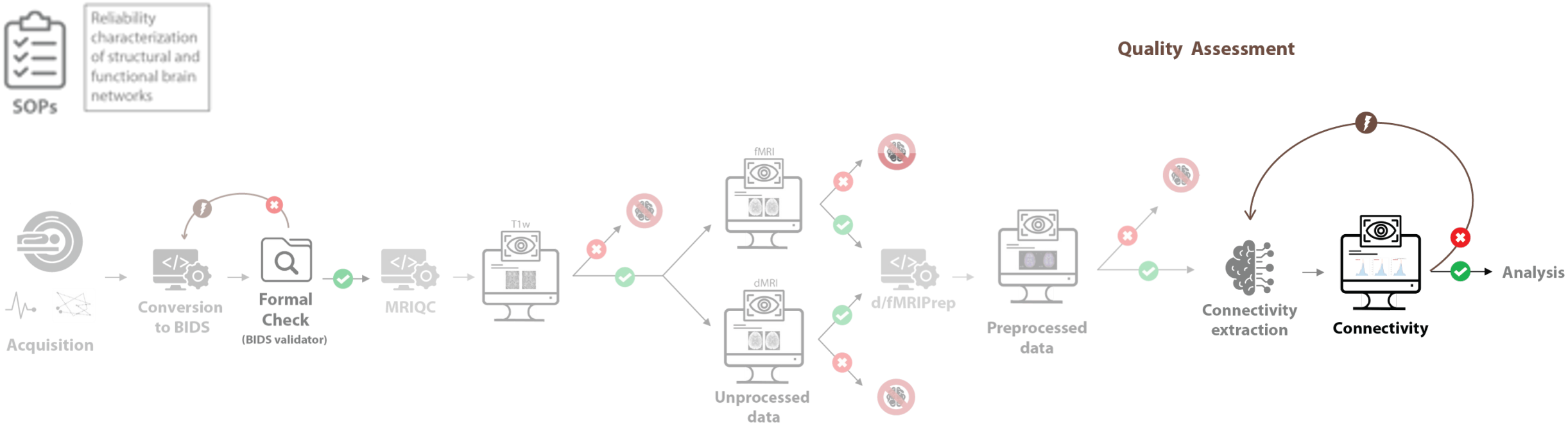
Quality Methods & Approaches for Multi-Site Studies

Tuesday, May 7 at 1.42pm

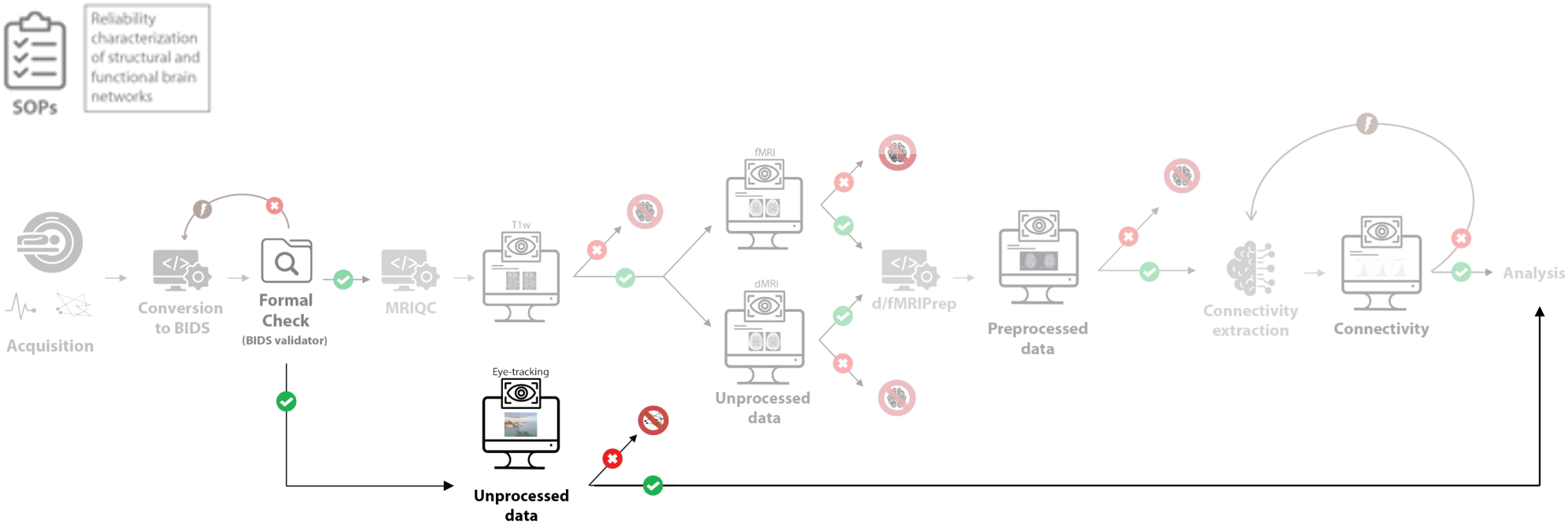
[12] Esteban et al. 2017, PLoS ONE

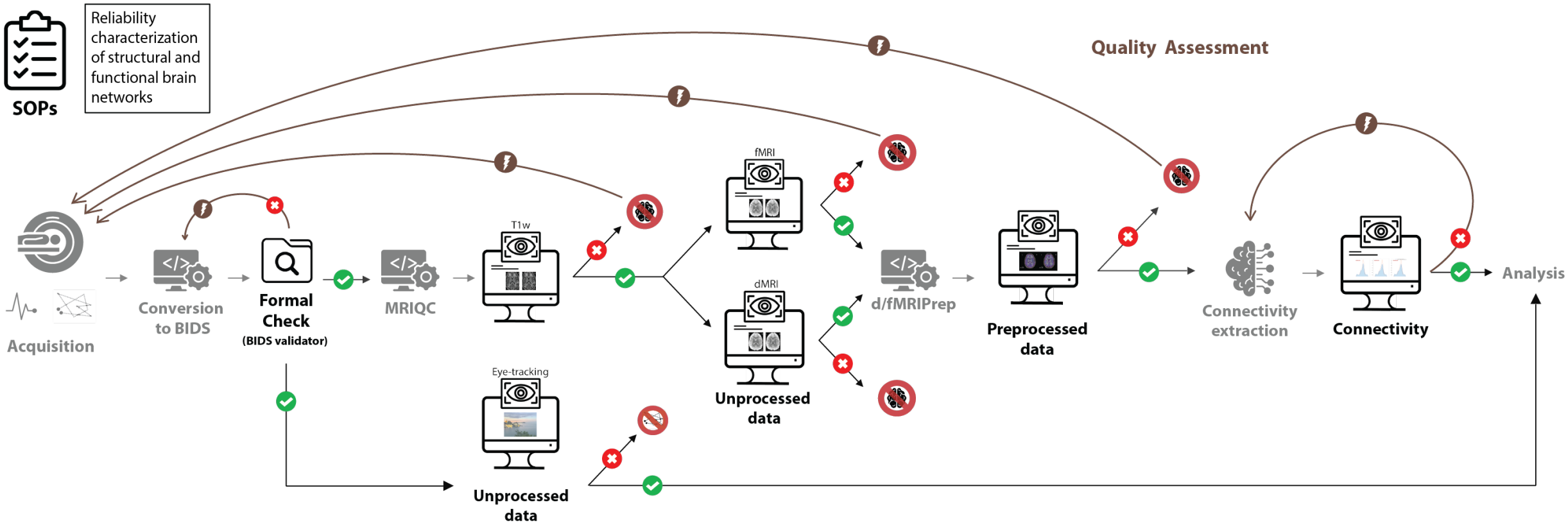


[13] Esteban et al. 2019, Nat. Meth.

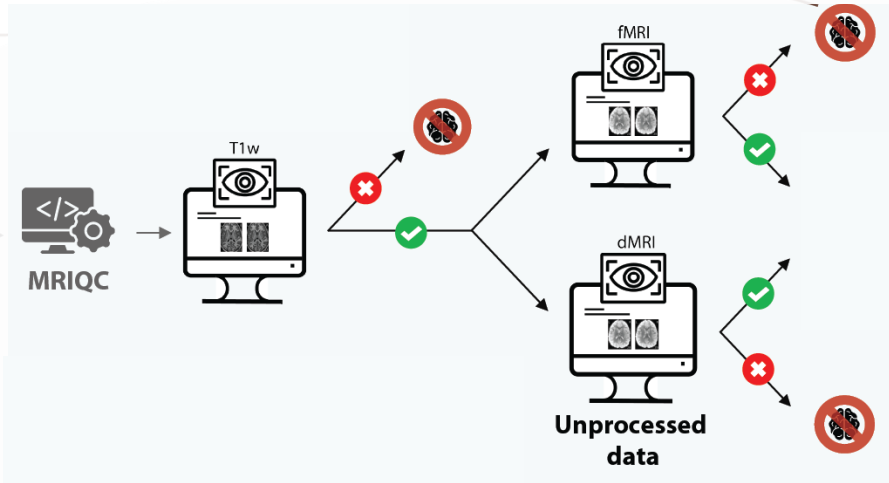


[14] Morfini et al. 2023, Front. Neurosci.
[15] Ciric et al. 2017, NeuroImage

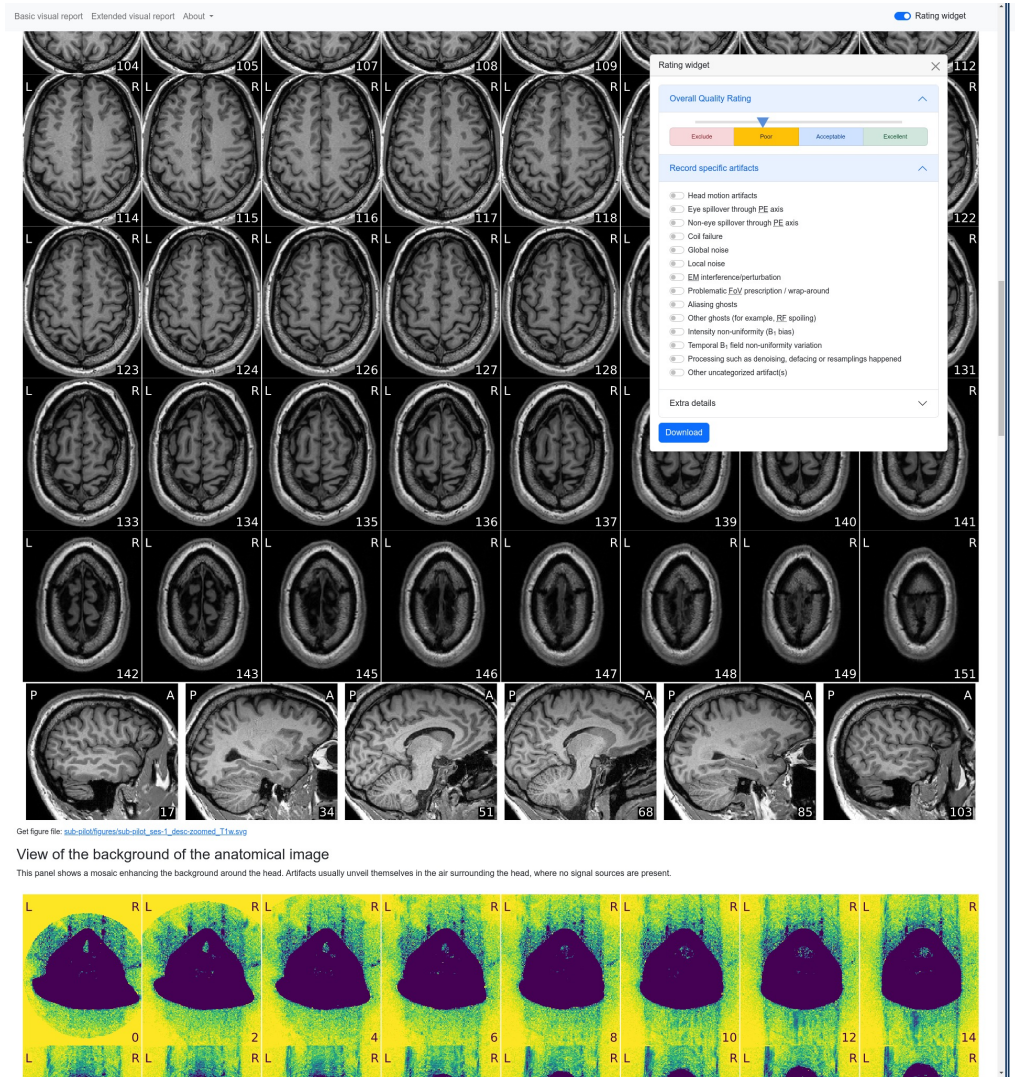

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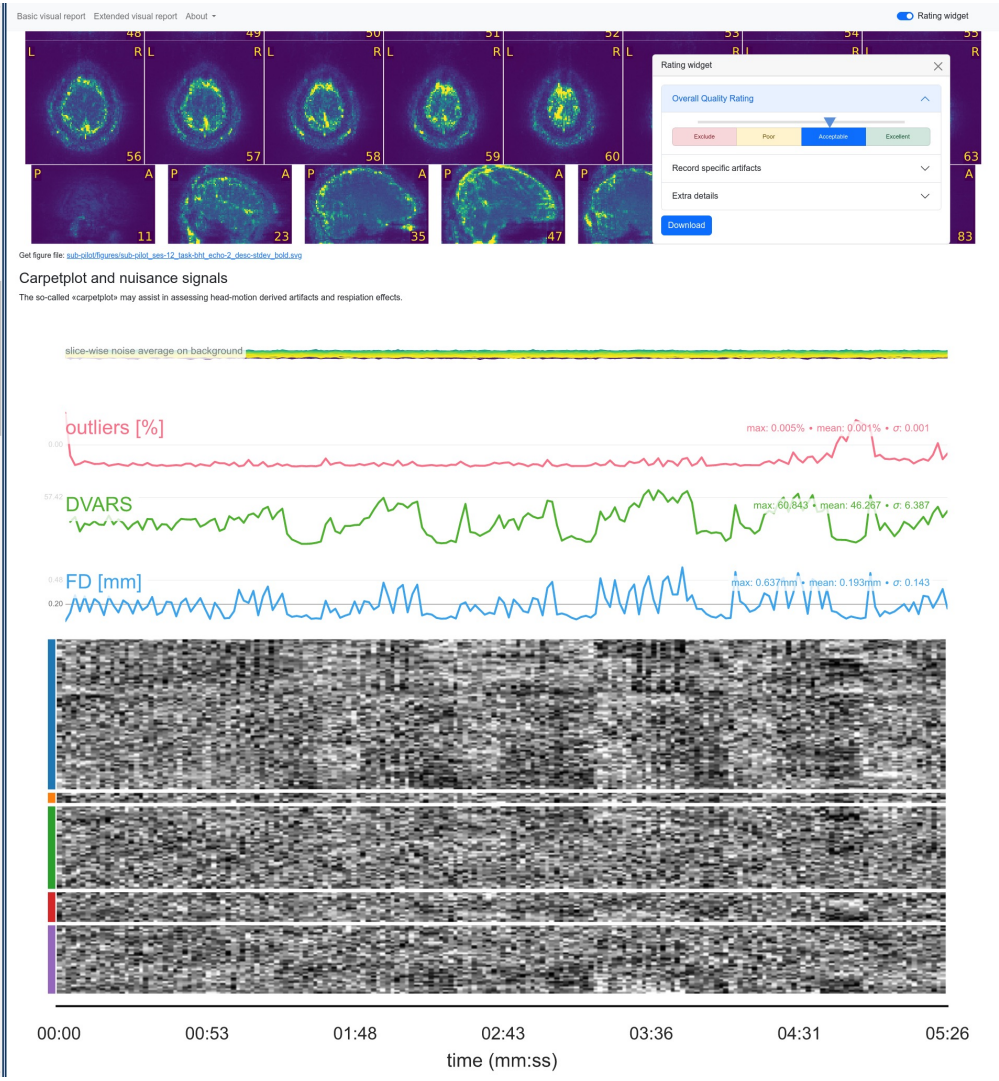


- QC checkpoints finalized for 3 modalities (T1w, fMRI and eye-tracking).


T1w
fMRI
dMRI
MRIQC
Unprocessed
data

MRIQC reports annotated by two experts





- All MRIQC reports and annotations for the HCPh dataset are available publicly at https://github.com/TheAxonLab/hcph-derivatives-mriqc

Regressor calibrated to flag HCPh T1w images for exclusion

Methods

- Regression predictive model implemented with **XGBRegressor**
- Optimization of the hyperparameters via **grid search**
- **3 repetitions of 10-fold cross-validation**
- Report performance with **Mean Absolute Error (MAE)**
- Baseline MAE using **dummy models** predicting average and median of target values

Regressor calibrated to flag HCPh T1w images for exclusion

Results

	XGBRegressor	DummyRegressor using mean	DummyRegressor using median
Mean MAE (std)	0.381 (0.098)	0.741 (0.164)	0.733 (0.165)

Future work

- Use the regressor to assist us in rating the T1w images from the second wave of HCPh acquisition.
- Evaluate the performance of the regressor on scans from the same brain but different scanners.
- Develop QA/QC protocol for physiological signals

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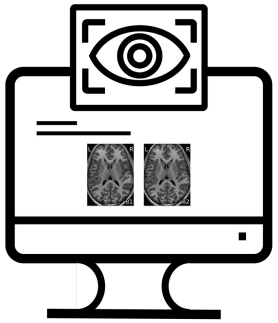
Acknowledgments

This work has been supported by the NIMH (RF1MH121867; OE) and the CZI/EOSS5 program ("NiPreps"; OE). CP, and OE receive support from the Swiss National Science Foundation —SNSF— (#185872; OE).

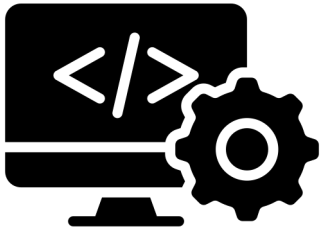
CONCLUSION



For more details about each QA/QC checkpoint, check our SOPs



All reports and expert ratings are publicly available



Notebook where we calibrated the regressor for QC of T1w images

Thank you for your attention!

Meet Oscar at 15:45
on Tuesday 7 May at the
Multi-Center Reproducibility & Tools
posters session, computer #83
if you have some questions,
suggestions or ideas.

Appendix

	Rated by	Number of images	Total rating time	Average time per image	Average rating ($\pm$std)	Number of excluded
T1w	CP	75	2h 16min	1min 49s	2.37 (poor) $\pm$ 0.85	11 (only pilots)
Multi-echo BOLD	CP	130	8h 58min	4min 8s	2.65 (acceptable) $\pm$ 0.49	1
DWI	OE	67	1h 21min	1min 13s	2.71 (acceptable) $\pm$ 0.38	1 pilot